Experiment # 8

Amplifier Frequency Response (BJT)

Objectives

- To obtain the frequency response of a BJT amplifier circuit.
- To determine the lower cutoff frequency and the upper cutoff frequency of a BJT amplifier circuit.
- To determine the midband gain, A_{mid} of a BJT amplifier circuit.
- To determine the input resistance, r_{in} and the output resistance, r_O of a BJT amplifier circuit.

List of Apparatus

Components

1. Transistor: BC109 (or equivalent general purpose npn)
2. Resistors: 150 KΩ, 33 KΩ, 4.7 KΩ, 680 Ω (2)
3. Capacitors : 0.22 μF (2) , 0.047 μF, 470 μF, 0.001 μF (2)

THEORY

The Frequency Response of an amplifier is presented in a form of a graph that shows output amplitude (or, more often, voltage gain) plotted versus frequency. Typical plot of the voltage gain of an amplifier versus frequency is shown in figure 1. The gain is null at zero frequency, then rises as frequency increases, level off for further increases in frequency, and then begins to drop again at high frequencies. The frequency response of an amplifier can be divided into three frequency regions.

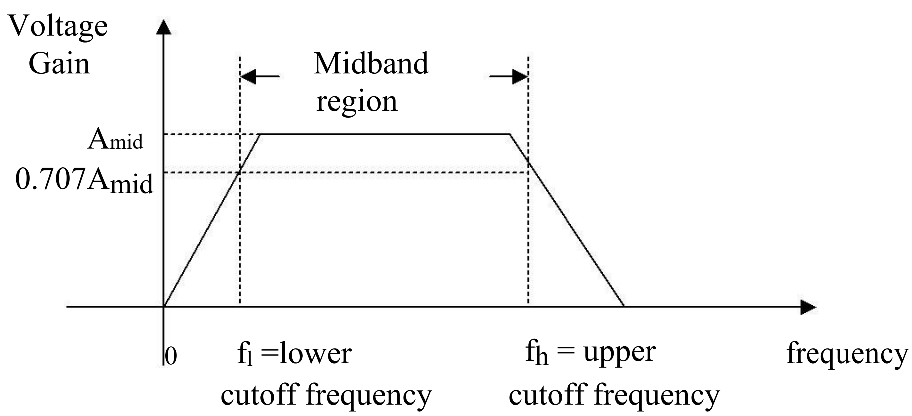


Figure 1. Frequency response of an amplifier.

The frequency response begins with the lower frequency region designated between 0 Hz and lower cutoff frequency. At lower cutoff frequency, f_L ,the gain is equal to 0.707 A_{mid}. A_{mid} is a constant midband gain obtained from the midband frequency region. The third, the upper frequency region covers frequency between upper cutoff frequency and above. Similarly, at upper cutoff frequency, f_H, the gain is equal to 0.707 A_{mid}. After the upper cutoff frequency, the gain decreases with frequency increases and dies off eventually.

The Lower Frequency Response.

Since the impedance of coupling capacitors increases as frequency decreases, the voltage gain of a BJT amplifier decreases as frequency decreases. At very low frequencies, the capacitive reactance of the coupling capacitors may become large enough to drop some of the input voltage or output voltage. Also, the emitter-bypass capacitor may become large enough so that it no longer shorts the emitter resistor to ground. Approximately, the following equations can be used to determine the lower cutoff frequency of the amplifier, where the voltage gain drops 3 dB from its midband value (=0.707 times the midband A_{mid}):

2. $f_l = 1/ (2\pi r_{in}C_1)$

where,

f_l = lower cutoff frequency due to C_1

C1 = input coupling capacitance

Ri = input resistance of the amplifier

(2) $f_2 = 1/ (2\pi r_o C_2)$

where,

f_2 = lower cutoff frequency due to C_2

C_2 = output coupling capacitance

r_o = output resistance of the amplifier

Provided that f_l and f_2, are not close in value the actual lower cutoff frequency is approximately equal to the largest of the two.

The Upper Frequency Response

The capacitive reactance of a capacitor decreases as frequency increases. This can lead to problems for amplifiers used for high-frequency amplification. Transistor has inherent shunt capacitances between each pair of terminals. At high frequencies, these capacitances effectively short the ac signal voltage. For circuit in figure 2, the upper cutoff frequencies f_{h1} and f_{h2} are due to shunt capacitances C_{bc} and C_{be},

Now C_{bc} is further divided into two capacitances which are

C_{min} and $C_{mout.}$ These are actually the modified Miller-effect capacitances, which is the interelectrode feedback capacitance C_{cb} modified by the gain A_{mid} as follows:

$$C_{min} = (1 - A_{mid})\, C_{bc} \;,\; C_{mout} = (1 - 1/A_{mid})C_{bc}$$

where,

A_{mid} = the voltage gain from input-to-load midband.

Now the two upper cutoff frequencies are found by

$f_{h1} = 1/ (2\pi(Rs||R_B||\beta re^{/})\, C_{min} + C_{be})$
$f_{h2} = 1/ (2\pi(Rc||R_L)\, C_{mout})$

f_h = smaller of f_{h1} and f_{h2}

PART I PRELIMINARY WORK

You are required to produce a simulated frequency response, voltage gain versus frequency, of the amplifier circuit given in figure 2 by using MULTISIM or of any equivalent software. Print the response. Observe the simulated response and determine the following values; the midband gain, A_{mid}, the lower cut-off frequency, f_L and the upper cut-off frequency, f_H. Record all the values.

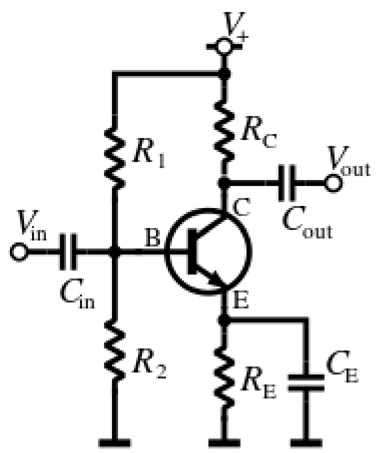
PART II. EXPERIMENTAL PROCEDURE

Setup your apparatus and do all the necessary connections as to carry out the required experimental work in determining the frequency response of the amplifier circuit as given in figure 2. Apply supply voltage to the circuit. Measure and record the base, the collector and the emitter voltage of the transistor. Next, apply a sinusoidal input signal, Vi, with vo1tage of 1volt peak-to-peak value at frequency of 5 Hz to the test circuit. Observe, measure, record the output voltage,V_O and calculate the voltage gain, Av = Vo/Vi. Also measure the phase difference between input and output signals and record in a table. With input signal always constant, increase signal frequency, measure and record the output voltage, V_O. Again calculate the voltage gain and obtain the signals phase difference. You are required to create a frequency response table to tabulate your readings with frequency, input voltage, output, voltage gain, phase difference and phase angle as the table entry list. Produce all necessary readings to cover frequency band between 5 Hz and 50 KHz. Make sure the input voltage is always constant at all frequency settings.

Plot the frequency response of (i) voltage gain in dB versus frequency, and (ii) phase response verses frequency on a semi-log paper. Explain and discuss your observation upon the simulated and the measured results.

QUESTIONS AND DISCUSSION

1. What are the simulated values of f_L, A_{mid} and f_H for the common-emitter amplifier in figure 2? Compare these values with the cutoff frequencies obtained from the measured values.
2. Which capacitor affects on the lower cutoff frequency and the upper cutoff frequency of the amplifier circuit?
3. With cutoff frequencies known, find the input resistance, r_{in}, and the output resistance, r_O , of the amplifier circuit.
4. Obtain the empirical function to describe the measured frequency response of the amplifier circuit?
5. Discuss your observation on the simulated and the measured frequency response.



R1 = 47 k, R2 =10 k, RC = 4.7k, RE = 900 Ohms, RF = 100 Ohms, Cin = 0.22 µF,

Cout = 0.22 µF, CE = 47 µF

Figure 2. Amplifier circuit for the frequency response experiment.0.

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